WORKSHEET 4.3 : SOLVING SYSTEMS OF EQUATIONS

a) $3x+2y-4z=-7$ ①
$2x-y+5z=13$ ②
$x-6y+2z=11$ ③

Eliminate x

① $3x+2y-4z=-7$
③ × 3. $3x-18y+6z=33$
⊖ $20y-10z=-40$.
$2y-z=-4$. ④

② $2x-y+5z=13$
③ × 2 $2x-12y+4z=22$
⊖ $11y+z=-9$ ⑤

Eliminate z

④ $2y-z=-4$
⑤ $11y+z=-9$.
⊕ $13y=-13$.
$y=-1$

sub $y=-1$ into ④
$2(-1)-z=-4$
$-z=-2$
$z=2$

sub into ③
$x-6(-1)+2(2)=11$
$x+6+4=11$
$x=1$

∴ P.O.I $(1,-1,2)$

b) $x+3y-z=3$ ①
$2x-4y+z=4$ ②
$3x+2y+2z=-4$ ③

Eliminate x

① × 2 $2x+6y-2z=6$
② $2x-4y+z=4$
⊖ $10y-3z=2$ ④

① × 3 $3x+9y-3z=9$
③ $3x+2y+2z=-4$
⊖ $7y-5z=13$ ⑤

Eliminate z

5 × ④ $50y-15z=10$
3 × ⑤ $21y-15z=39$.
⊖ $29y=-29$
$y=-1$

sub into ④
$10(-1)-3z=2$
$-3z=12$
$z=-4$

sub into ①
$x+3(-1)-(-4)=3$
$x+1=3$
$x=2$

∴ P.O.I $(2,-1,-4)$

c) $2x + 7y - z = 28$ ①

$3x - y + 2z = -10$ ②

$4x + y + 5z = -18$ ③

Eliminate y

② $3x - y + 2z = -10$

③ $4x + y + 5z = -18$

④ $7x + 7z = -28$

$x + z = -4$ ④

① $2x + 7y - z = 28$

③ × 7 $28x + 7y + 35z = -126$

⑤ $-26x - 36z = 154$

$13x + 18z = -77$ ⑤

Eliminate z

④ × 18 $18x + 18z = -72$

⑤ $13x + 18z = -77$

⑥ $5x = 5$

$x = 1$

Sub $x = 1$ into ④

$1 + z = -4$

$z = -5$

Sub into ②

$3(1) - y + 2(-5) = -10$

$-y - 7 = -10$

$-y = -3$

$y = 3$

∴ P.O.I $(1, 3, -5)$

d) $x + 2y - z = -5$ ①

$3x - y - 2z = 3$ ②

$4x + 3y + 3z = 18$ ③

Eliminate z

① × 2 $2x + 4y - 2z = -10$

② $3x - y - 2z = 3$

⊖ $-x + 5y = -13$ ④

① × 3 $3x + 6y - 3z = -15$

③ $4x + 3y + 3z = 18$

⊕ $7x + 9y = 3$ ⑤

Eliminate x

④ × 7 $-7x + 35y = -91$

⑤ $7x + 9y = 3$

⊕ $44y = -88$

$y = -2$

sub $y = -2$ into ④

$-x + 5(-2) = -13$

$-x = -3$

$x = 3$

sub into ①

$(3) + 2(-2) - z = -5$

$3 - 4 - z = -5$

$-z = -4$

$z = 4$

∴ P.O.I. $(3, -2, 4)$

e) $5x+y-z=-2$ ①

$2x+2y-3z=3$ ②

$x+3y-4z=7$ ③

Eliminate x

① $5x+y-z=-2$

③ ×5 $5x+15y-20z=35$

⊖ $-14y+19z=-37$ ④

② $2x+2y-3z=3$

③ ×2 $2x+6y-8z=14$

⊖ $-4y+5z=-11$ ⑤

Eliminate y

④ × 2 $-28y+38z=-74$

⑤ × 7 $-28y+35z=-77$

⊖ $3z=3$

$z=1$

Sub $z=1$ into ⑤

$-4y+5(1)=-11$

$-4y=-16$

$y=4$

Sub into ③

$x+3(4)-4(1)=7$

$x+8=7$

$x=-1$

∴ P.O.I $(-1, 4, 1)$